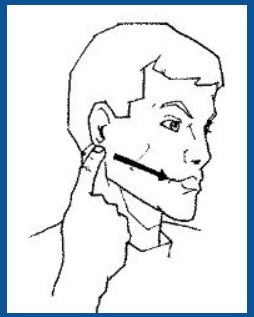


Silent Voice Bridge: The Edge AI ASL Interpreter

Team: Aramish Farooq, Pwint Thu Thu San, Reuben George, Nathaniel Ortega

Problem

Deaf and hard-of-hearing individuals face significant communication barriers. Existing translation solutions are inaccessible, require expensive proprietary hardware, or rely heavily on cloud processing, which raises privacy concerns and latency issues.



Solution

A software-only, offline Edge-AI platform. We transform a standard webcam into a real-time ASL interpreter, bringing secure, low-latency bidirectional communication directly to edge devices like the NVIDIA Jetson.

195A Accomplishments & Proof of Concept

The Transition: From Static Alphabet to Dynamic Words

- Evolved from single-frame letter classification (A–Z) to multi-frame temporal recognition of continuous gestures and words.

The Challenge: "Movement Epenthesis"

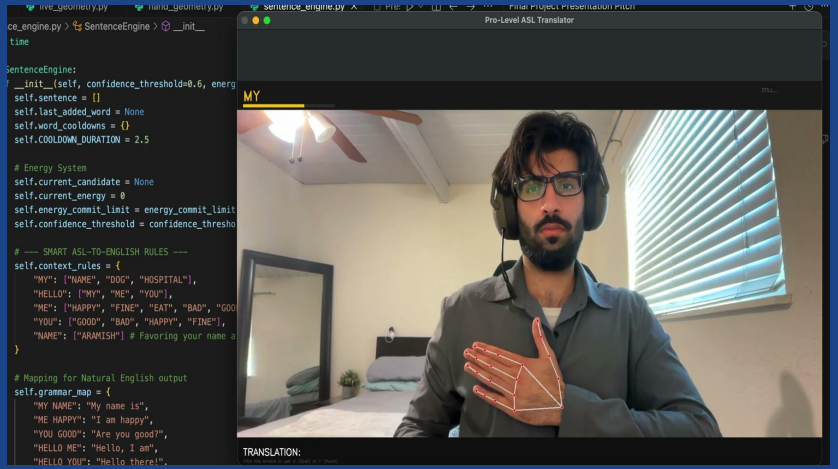
- Moving hands between signs generated messy transitional noise, causing random false predictions and rapid text flickering.

The Fix: Temporal Smoothing & Energy Commit Bar

- Implemented a 30-frame sliding window combined with an "energy-based" confidence threshold.
- The system ignores high-velocity hand movement and only triggers an output once the sign is clearly held.

The Result: Production-Grade Stability

- Eliminated jitter and false triggers, delivering clean, stable real-time text output at 30+ FPS without needing specialized hardware.



System Architecture & Technical Learnings

Algorithm Evolution: Pivoted from our LSTM baseline to a Spatial-Temporal Graph Convolutional Network (ST-GCN). Graph convolutions proved significantly superior at mapping the physical topology of finger joints over time.
Data Learnings: Discovered that model generalization relies entirely on dataset diversity. We integrated the ASL Citizen dataset and applied CLAHE lighting normalization to handle real-world environmental variables.



All five stages run locally on the NVIDIA Jetson, so no frames leave the device



195B Implementation Plan & Deliverables

Vocabulary & Phrase Expansion: Scaling our dataset to include continuous phrases and a vastly expanded dictionary of target words. We are actively training the ST-GCN to handle these complex multi-sign sequences.

Edge Hardware Optimization: Rewriting and compiling our algorithms to fit strictly within the memory and compute limits of the NVIDIA Jetson target hardware.

Gemma VLM Exploration: Developing a parallel experimental pipeline using the Gemma Vision-Language Model. We are testing its capability to process raw video feeds and act as an intelligent semantic grammar engine to structure translated words into natural English sentences.



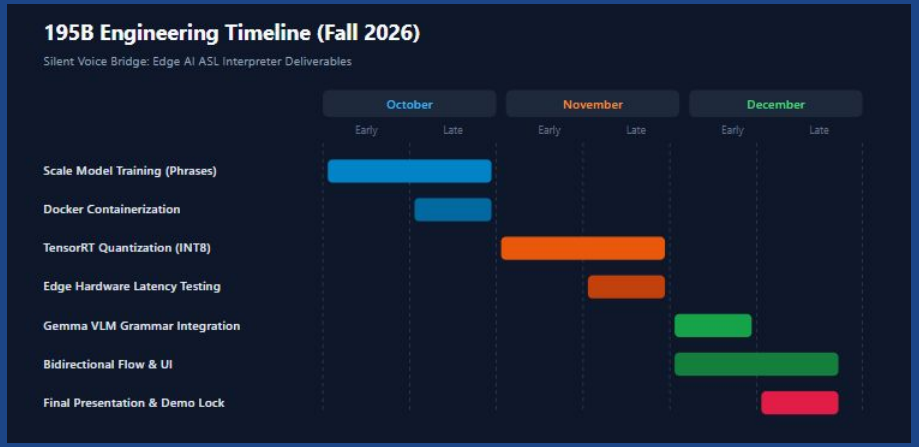


Semester Timeline

October: Scale model training for extended vocabulary/phrases and finalize Docker containerization for the Jetson deployment.

November: Apply NVIDIA TensorRT optimizations to compress the expanded model, ensuring it fits Jetson limits while maintaining sub-300ms latency.

December: Complete Gemma VLM integration tests and finalize the bidirectional (Hearing-to-DHH) communication flow for the final presentation demo.



Implementation Risks & Mitigations

Hardware Limits (Jetson 4GB)

- *Risk:* Larger vocabulary could exceed memory, throttling FPS.
- *Mitigation:* Apply TensorRT INT8 quantization and optimize queue buffers.

VLM Inference Latency

- *Risk:* Running Gemma locally alongside vision causes conversational lag.
- *Mitigation:* Decouple pipelines; use ST-GCN for live tracking and Gemma asynchronously for grammar.

Sign Ambiguity & Confusion

- *Risk:* Expanded vocabulary increases overlap between visually similar signs.
- *Mitigation:* Add heuristic prediction cooldowns and target-augment similar sign data.

